ronmental conditions and in various atmospheres. These can provide temperature measurements in the range of -270°C to 1260°C, and an output of -6.4mV to 54.9mV over maximum temperature range. Thermal response of K-type thermocouples is also fast as compared to RTDs and thermistors.

Circuit and working

Fig. 3 shows the complete circuit diagram of the system. K-type thermocouple is used for temperature sensing and MAX6675 module is used as signal converter. MAX6675 performs cold-junction compensation and digitises the signal from K-type thermocouple. Data is output in a 12-bit resolution and communicated to the MCU through SPI protocol. MAX6675 provides a resolution of 0.25°C, and allows temperature readings from 0°C to +1024°C.

MSP430G2553 (IC2) is an ultra-low-power mixed-signal MCU with 16-bit RISC CPU, and has a digitally-controlled oscillator for data processing. A twenty-pin MSP430G2553 MCU is used here. It periodically acquires data from the signal converter and converts the digital data into corresponding temperature value.

Converted values are sent to the four-digit, seven-segment display (DIS1) through an analogue multiplexer/demultiplexer HEF4051B (IC3). Through pin P2.7 of MCU, temperature data is sent serially to pin $\bar{E}$ of HEF4051B to update the segments of the digit.

Pins P2.0, P2.1 and P2.2 of the MCU are connected to selection pins S1, S2 and S3 of HEF4051, respectively, to select the particular segment of a digit in DIS1. Pins P2.3, P2.4, P2.5 and P2.6 of the MCU are connected to pins 12, 9, 8 and 6 of DIS1, respectively, to select a particular digit.

The MCU also sends data to the smartphone through HC-05 module. HC-05 is a Bluetooth Serial Port Protocol (SPP) module interfaced with the MCU through UART. Pins RX and TX of HC-05 are connected to pins P1.1 and P1.2 of the MCU, respectively. HC-05 Bluetooth module is used to communicate between the smartphone and the circuit. A 9V PP3 battery or a suitable DC power supply adaptor is used to power up the system. LM317 is used as voltage regulator to provide 3.3V to 3.6V.

DIS1 is capable of displaying temperature up to 999.8°C. Temperature measurement can also be viewed through an Android smartphone with Bluetooth Terminal HC-05 app installed. Connect Bluetooth of the smartphone with HC-05 Bluetooth module in the circuit. Default auto-pairing password is 1234.

Next, open the app; you will notice a list of nearby Bluetooth devices. Select HC-05 from the list to connect it.

Send ST ASCII value to HC-05. Whenever UART of the MCU gets the command ST, it will send data to HC-05, as shown in Fig. 4.

Fig. 4: Output through Bluetooth terminal on smartphone

Software

Fig. 5 shows the integrated development environment (IDE) of the IAR-embedded workbench for MSP430 version 7.12. KickStart edition is used to develop the firmware in embedded C and program MSP430G2553. MSP430 LaunchPad Kit is used to burn the program to the MCU.

The following steps are used to develop and burn the firmware to the MCU:

1. Install IAR-embedded workbench for MSP430 (IAREW) on Windows PC.
2. Register for KickStart (free) licence.
3. Open IAR-embedded workbench IDE.
4. Select File→New Workspace.
5. Open Project→Create New Project→Expand C Tree Node→Main, Toolchain→MSP430. Click OK.
6. Give a filename and save it.
7. Copy and paste the source code (main.c) in the text editor.
8. On the workspace (left pane), right-click on the project name. Select Options→Category→General Options Device→MSP430G2553.
9. Select Category→Debugger→Driver→FET Debugger.

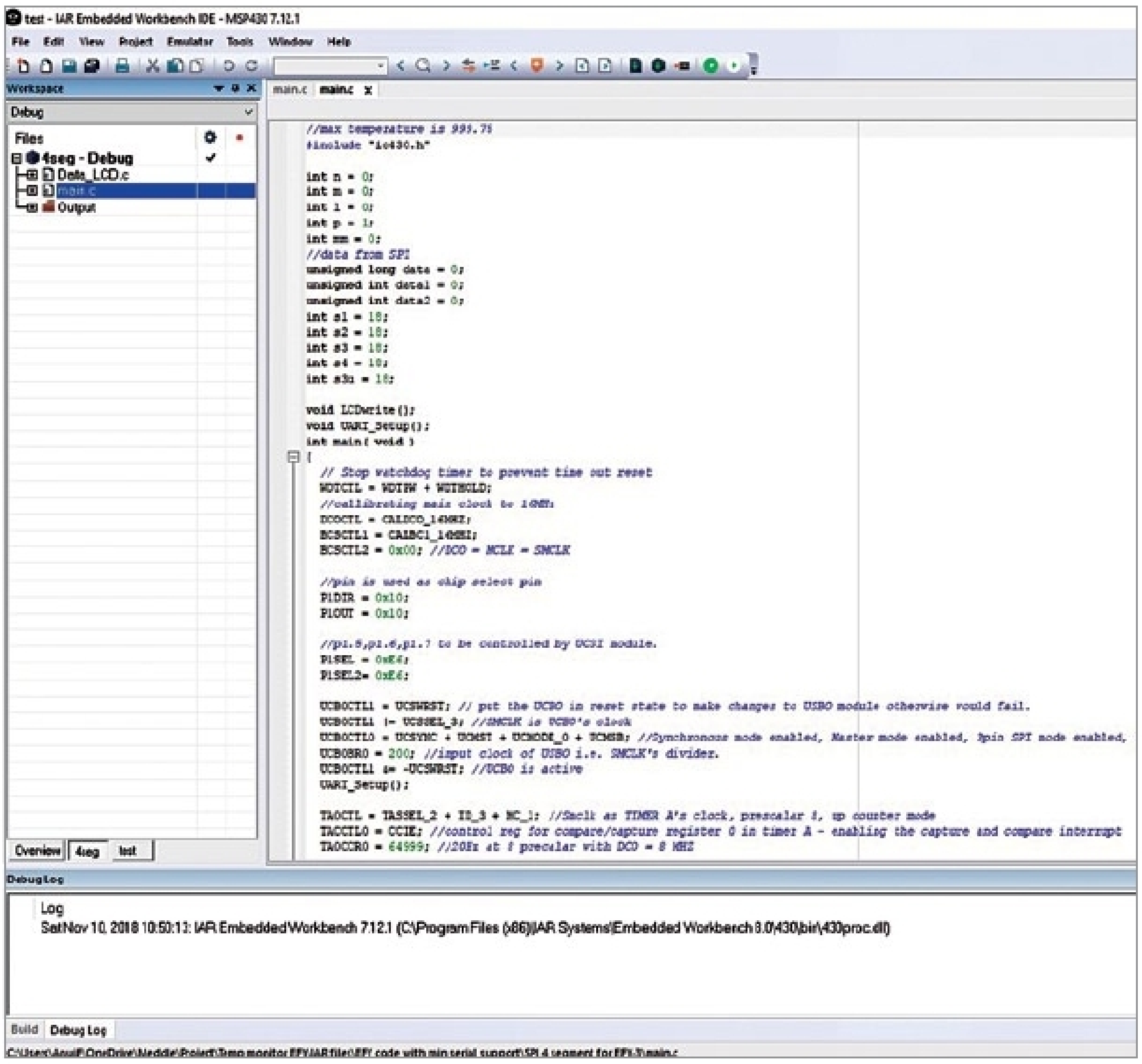

Fig. 5: IDE of IAR-embedded workbench